```
#dplyr
install.packages("dplyr")
library(dplyr)
data("storms")
#select: extraction by column
s1=select(storms, status, wind)
s2=select(storms,-status,-wind)
s3=select(storms,name:day)
s4=select(storms,2:4)
s5=select(storms,1,8,10,11)
s6=select(storms, contains("force"))


#filter: extraction by row
f1=head(storms,10)
f2=filter(f1,wind>=30)
f3=filter(f1,wind>=30,status=="tropical storm")


#using pipeoperator
f4=filter(f1, wind      >=      30, status      %in%    c("tropical storm"))

#for calculations from existing columns using mutate
m1=mutate(f1, ratio = pressure / wind)
m2=slice(f1,2:6)


A=c(1,3,5)
m=min(A)
avg=mean(A)
#pointwise min of columns
a=data.frame(pmin(storms$month, storms$hour))

#summarise: to change the unit (big data to small values)
Avg_mpg=summarise(mtcars,avg=mean(mpg))
auto_yn=group_by(mtcars,am)
auto_yn1=summarise(auto_yn,avg=mean(mpg))

#pipe operator use "ctr+shft+m" for this operator
New= group_by(mtcars,am) %>% summarise(avg=mean(mpg))

f1 %>% summarise(average = mean(hour), total = sum(hour), count = n())

n_distinct(f1$hour) #The number of distinct values in a vector

#arrange in ascending or descending
B=arrange(storms,pressure)
C=arrange(storms,desc(pressure))


View(mtcars)
#extract by variables i. e column
select
```